

SCTracker

Native Apps · Benutzerhandbuch · User Manual

Deutsch · English · አማርኛ · ትግርኛ

Coffee only · Product prototype · September 2026

SCTracker Native Apps — Benutzerhandbuch

Deutsch · Referenzimplementierung

Version 0.1 — September 2026

WICHTIG: Die iOS- und Android-Apps sind ausführbare Referenzimplementierungen, keine Produktionssysteme, Sicherheitszertifizierungen, Rechtsberatung oder automatische EUDR-Konformitätsentscheidung.

1. Zweck und Sicherheitsgrenzen

SCTracker erfasst offline Nachweise zu Kaffeechargen, Säcken, Siegeln, Übergaben, Standort, Medienreferenzen und Paketprüfung. Ereignisse werden nur angehängt und per Hash verkettet, damit spätere Änderungen erkennbar sind.

Die Referenz zeigt lokale Signaturen, Lebenszyklusregeln, Übergabeentscheidungen, Quarantäne und geprüften Paketaustausch. Für Produktion sind zusätzlich Backend-Identität und -Autorisierung, Organisationsvertrauen und Widerruf, sichere Registrierung und Wiederherstellung, verschlüsselte Sicherung und Übertragung, Aufbewahrungsregeln, Überwachung, Datenschutzprüfung, unabhängige Protokoll-/Sicherheitsprüfung und geschulte Bediener erforderlich. Ein Hash oder eine Gerätesignatur beweist weder die Wahrheit einer Angabe noch die Berechtigung eines Akteurs.

2. Installation und Einrichtung

Android

1. Android 8.0/API 26 oder neuer mit Displaysperre verwenden; nur einen von der Organisation freigegebenen, signierten Build installieren.
2. Für Entwicklung `mobile-native\android` mit JDK 17 und Android SDK 35 in Android Studio öffnen oder `.\gradlew.bat assembleDebug` ausführen.
3. Nur erforderliche Kamera-, Standort-, Datei- und NFC-Berechtigungen erteilen. Die Referenz modelliert Medien und GPS, enthält aber noch nicht alle Erfassungsabläufe.
4. Prüfen, dass der Header „offline bereit“ meldet und das Sprachsymbol dauerhaft sichtbar ist.

iOS

1. iOS 17 oder neuer mit Gerätecode verwenden; nur freigegebene signierte Builds installieren.
2. Für Entwicklung `mobile-native\ios\SCTrackerNative.xcodeproj` in Xcode 16 oder neuer öffnen und das Schema `SCTrackerNative` wählen.
3. NFC auf einem Gerät benötigt die richtige App-ID und NFC-Tag-Reading-Berechtigung. Der Simulator nutzt nur einen Entwicklungs-Softwareschlüssel.
4. Prüfen, dass das Globus-Symbol in jedem Bildschirm-Header sichtbar bleibt.

Simulator-Daten, Entwicklungsschlüssel, Debug-Builds und Beispielakteure dürfen nicht für Produktionsnachweise verwendet werden.

3. Geräteregistrierung und Vertrauenspaket

Mitgelieferte Akteure, Rollen und Vertrauensstatus sind Platzhalter. In Produktion registriert ein Administrator das Gerät, bindet den öffentlichen Schlüssel an Organisation und benannten Bediener, dokumentiert Hardware/OS/App und Schlüsselschutz, stellt ein signiertes Vertrauenspaket mit Rollen und Ablauf aus und ermöglicht Widerruf.

Vor Arbeitsbeginn Geräte-ID, Schlüssel-ID, Organisation, Bediener, Rollen, Gültigkeit und Signatur des Vertrauenspakets mit dem Einsatzauftrag abgleichen. `UNVERIFIED`, `LOCALLY_TRUSTED`, `ORGANIZATION_VERIFIED` und `REVOKED` sind Protokollkennungen; nur die Organisationsrichtlinie vergibt Befugnisse. Abgelaufene, unbekannte, unpassende oder widerrufene Pakete ablehnen.

4. Sprachauswahl

Das dauerhafte Globus-/Sprachsymbol im oberen Header antippen und Deutsch, English, አማርኛ oder ትግርኛ wählen. Die Auswahl wird lokal gespeichert. Kennungen, Hashes, Codes, Eigennamen und Protokollwerte bleiben unverändert; erklärende Beschriftungen werden übersetzt.

5. Offline-Vorbereitung

- Gerät laden und genügend geschützten Speicher prüfen.
- App öffnen, Sprache wählen und Bediener-/Gerätevertrauen prüfen.
- In Produktion Arbeitsauftrag und freigegebenes Vertrauenspaket herunterladen.
- Datum/Uhrzeit, Kamera, Standort und Berechtigungen prüfen.
- Sicheren Paketübertragungsweg und Papier-Ersatzverfahren vorbereiten.
- Nicht auf NFC verlassen: echte Chipvorgänge sind absichtlich gesperrt.

„Offline bereit“ bedeutet nur, dass lokale Abläufe funktionieren; Identität, Vertrauen, Sicherung und Synchronisierung sind damit nicht abgeschlossen.

6. Bedienerauthentifizierung

Die Referenz enthält keine Produktionsanmeldung. Produktion muss Organisationsidentität, Geräteentsperrung, kurze Inaktivitätssperre, rollenbasierte Autorisierung jeder Aktion und Aufsicht für sensible Korrekturen/Abgleiche erzwingen. Zugangsdaten nie teilen und entsperrte Geräte nie unbeaufsichtigt lassen. Bei falschem Bediener, falscher Organisation oder Rolle Arbeit stoppen.

7. Genesis und Erfassung von bis zu 100 Säcken

1. Abläufe/Säcke öffnen und neue Chargen-Genesis wählen.
2. Chargenreferenz und Herkunft exakt aus den Quelldokumenten übernehmen.
3. Sackzahl von 1 bis 100 einstellen; andere Werte werden abgelehnt.
4. Charge erstellen. Die App erzeugt deterministische Sackreferenzen und startet mit `UNISSUED`.
5. Jede angezeigte Sackreferenz mit der physischen Kennzeichnung vergleichen.

Kennungen nie wiederverwenden und Genesis nie still ändern. Fehler werden nach Produktionsrichtlinie durch ein autorisiertes Korrektur- oder Ungültigkeitsereignis berichtigt.

8. GPS-Qualität

Standort im Freien bei freier Sicht erfassen. Auf stabilen Fix warten und horizontale Genauigkeit, Quelle, Erfassungszeit sowie gegebenenfalls Mock-Standort prüfen. Die Referenz markiert höchstens 25 Meter als akzeptabel; Einsatzregeln können strenger sein. Schlechte Messung wiederholen und fehlenden Fix begründen. GPS ist Nachweismetadatum, kein Herkunftsbeweis; Koordinaten nie erfinden, runden oder von einem anderen Ort kopieren.

9. Lebenszyklus, Kontrollen, Schaden, Öffnung und Korrekturen

Normal: UNISSUED ISSUED SEALED IN_TRANSIT OPENED. VOID und DAMAGED sind terminale Ausnahmezustände. Die Oberfläche bietet nur erlaubte Aktionen.

- Erst nach Prüfung von Genesis und Säcken ausgeben.
- Nur nach freigegebenem Siegelverfahren versiegeln.
- Transport erst mit korrekten Übergabe-/Versanddaten starten.
- Nur am autorisierten Ziel öffnen und Nachweis erfassen.
- Schäden an Sack oder Siegel als beschädigt markieren, normale Bearbeitung stoppen und Quarantäne befolgen.
- Nicht nutzbare Kennungen als ungültig markieren.
- Historie nie überschreiben; Korrekturen sind neue signierte Ereignisse mit Bezug zum Fehler und erforderlicher Berechtigung.

10. OFFER, ACCEPT, REJECT und PENDING

OFFER erzeugt eine Übergabe im Status PENDING; Verwahrung ist noch nicht übertragen. Der Empfänger prüft Chargen-/Sackreferenzen, Sender, Empfänger, Zustand und exakten Angebots-Hash. ACCEPT oder REJECT muss exakt diesen Hash referenzieren. Zweite Entscheidungen und falsche Hashes werden abgelehnt. Streitige oder parallele Übergaben bleiben bis zur Aufsichtsprüfung ausstehend/quarantänisiert; nicht neu erstellen, um Konflikte zu verbergen.

11. Foto- und Dokumentnachweise

Etiketten, Siegel, Zustand und zulässige Dokumente scharf und vollständig fotografieren. Unbeteiligte Personen, Ausweise, Bildschirme und private Umgebung vermeiden. Fokus, Vollständigkeit, Zeit, Chargenzuordnung, MIME-Typ, Größe und SHA-256-Referenz prüfen. Die Referenz speichert Modelle/Hashes, keinen vollständigen sicheren Medientresor. Produktion benötigt verschlüsselte Speicherung, Upload-Wiederholung, Malware-Prüfung, Aufbewahrung/Löschung und Zugriffskontrolle.

12. Quarantäne, Konflikte und Prüfung

Charge oder Paket quarantänisieren, wenn Signatur, Hash, Sequenz, Vorgänger-Hash, Identität/Vertrauen, Lebenszyklus, Nachweis oder physischer Zustand nicht geprüft werden kann. Abweichende Ereigniszweige werden nicht automatisch zusammengeführt. Beide Zweige erhalten, Fehlercode notieren, weitere Aktionen stoppen und eskalieren. „Als geprüft markieren“ dokumentiert in der Referenz nur die Prüfung; Produktion benötigt eine signierte Ausgleichsentscheidung durch autorisierte Aufsicht.

13. Paketexport, -import und Prüfung

Für Export „signierten Export erstellen“ wählen, Paket unverändert lassen und über freigegebenen Kanal übertragen. Für Import „importieren und prüfen“ wählen. Vor Annahme werden Manifeststruktur/-signatur, exakte Dateimege, sichere Pfade, Bytegrößen, alle SHA-256-Hashes, Kettenwurzel, Ereignisketten und Signaturen geprüft.

Nie teilweise importieren, Inhalte umbenennen/ändern, Prüfungsfehler umgehen oder eine eigenständige Signatur als Organisationsvertrauen behandeln. Ungültige/abweichende Pakete bleiben in Quarantäne. Produktion benötigt außerdem Größenlimits, Streaming-Archive, Empfänger verschlüsselung, Malware-/Medienprüfung, vertrauenswürdige Schlüsselsuche, Replay-Schutz und auditierte Synchronisierung.

14. Fehlercodes und Reaktion

- SCT-INPUT / GENESIS_SACK_COUNT_INVALID: Pflichtangaben oder Sackzahl 1–100 korrigieren.
- SCT-STATE / TRANSITION_NOT_ALLOWED: Datensatz aktualisieren und nur erlaubte Aktion ausführen.
- SCT-EVENT: stoppen und Daten erhalten; Duplikat, Sequenz, Hash oder Kette ist fehlerhaft.
- SCT-SIGN / KEYSTORE_UNAVAILABLE: Signieren stoppen; Registrierung und Schlüsselschutz prüfen.
- SCT-PACKAGE / MANIFEST_*: nicht importieren; Paket quarantänisieren und Hash melden.
- SCT-CONFLICT / IMPORT_QUARANTINED: Chargenarbeit stoppen und autorisierten Abgleich anfordern.
- SCT-STORAGE: Arbeit stoppen, Gerät sichern, Support kontaktieren; App-Daten nicht löschen.
- SCT-CHIP / SCT-NFC: nicht als Echtheitsprüfung wiederholen; freigegebenen Nicht-NFC-Ersatz nutzen.

Exakten Code, Zeitpunkt, Geräte-ID, Chargen-/Übergabe-/Paketkennung und Aktion notieren. Keine Geheimnisse, vollständigen Personendaten, APDUs oder Chipantworten an Support senden.

15. Datenschutz und verlorenes Gerät

Nur notwendige Daten erfassen. Passwörter, Schlüssel, unnötige Personendaten, vollständige Dokumentbilder oder genaue Koordinaten nie in Notizen, Logs, Chat oder öffentliche Systeme schreiben. Einwilligung, Zugriff, Aufbewahrung und Löschung beachten.

Bei Verlust/Diebstahl sofort melden, Geräteschlüssel und Vertrauenspaket widerrufen, Sitzungen sperren, letzten vertrauenswürdigen Ereignis-/Paketcheckpoint ermitteln, spätere nicht vertrauenswürdige Aktivität quarantänisieren und nur nach Richtlinie sperren/löschen. Ersatzgerät neu registrieren; Signierschlüssel nie aus App-Backup kopieren.

16. Fehlerbehebung

- App startet nicht: OS, Speicher, Signatur und Geräterichtlinie prüfen; vor Beweissicherung nicht neu installieren.
- Falsche Sprache: dauerhaftes Globus-Symbol nutzen; nur neu starten, wenn Auswahl nicht übernommen wird.
- Kein Standort: ins Freie gehen, Berechtigung aktivieren, Genauigkeit abwarten und Ersatz dokumentieren.
- Import scheitert: Originalpaket unverändert aufbewahren; Hash und Code melden.
- Konflikt: Betrieb stoppen und quarantänisieren; nie manuell einen Zweig auswählen.
- NFC scheitert: beim ausgelieferten deaktivierten Profil erwartet; Ersatzverfahren nutzen.
- Signatur/Speicher scheitert: Erfassung stoppen; App-Daten nicht löschen und Gerätezeit nicht ändern.

17. Verbindliche NFC-Warnung

ECHTE NFC-AUTHENTIFIZIERUNG, LESEN, AKTUALISIEREN UND RÜCKLESEN MÜSSEN GESPERRT BLEIBEN, bis ein genehmigter herstellerspezifischer PoC Chip und Speicher-/App-Layout identifiziert, autoritative APDUs/Statuswörter nutzt, sichere nicht in der App eingebettete Schlüsselbereitstellung, Diversifizierung, Rotation und Widerruf definiert, iOS-/Android-Kompatibilität belegt, Replay, Relay, Klonen, Tearing, Unterbrechung, Sperre, Schaden und Rücklesen testet und unabhängig sicherheitsgeprüft ist.

Das Repository enthält kein freigegebenes Produktions-Chipprofil, APDU-Set oder Bereitstellungsschlüssel. UID dient nur zur Suche und ist nie Echtheitsnachweis.

SCTracker Native Apps — User Manual

English · Reference implementation

Version 0.1 — September 2026

IMPORTANT: The iOS and Android apps are executable reference implementations, not production systems, security certifications, legal advice, or an automatic EUDR compliance decision.

1. Purpose and security limits

SCTracker records coffee batch, sack, seal, custody-transfer, location, media-reference, and package-verification evidence while a device is offline. Events are append-only and hash-linked so later changes are detectable.

The reference apps demonstrate local signing, lifecycle rules, transfer decisions, quarantine, and verified package exchange. Production deployment additionally requires backend identity and authorization, organization trust and revocation, secure enrollment and recovery, encrypted backup and transport, approved retention, monitoring, privacy review, independent protocol/security review, and trained operators. A hash or device signature proves neither that a fact is true nor that an actor was authorized.

2. Installation and setup

Android

1. Use Android 8.0/API 26 or newer with a screen lock; install only an organization-approved signed build.
2. For development, open `mobile-native\android` in Android Studio with JDK 17 and Android SDK 35, or run `.\gradlew.bat assembleDebug`.
3. Grant only required camera, location, file, and NFC permissions. The current reference UI models media and GPS data but does not complete all permission/capture flows.
4. Confirm the header says the device is offline-ready and that the language icon remains visible.

iOS

1. Use iOS 17 or newer with a passcode; install only an organization-approved signed build.
2. For development, open `mobile-native\ios\SCTrackerNative.xcodeproj` in Xcode 16 or newer and select the `SCTrackerNative` scheme.
3. Physical-device NFC requires the correct App ID and Near Field Communication Tag Reading entitlement. The simulator uses a development-only software signing key.
4. Confirm the globe icon remains visible in every screen header.

Do not use simulator data, development keys, debug builds, or sample actors for production records.

3. Device registration and trust package

The shipped actors, roles, and trust states are placeholders. A production administrator must register the device, bind its public key to an organization and named operator, record hardware/OS/app versions and key-protection level, issue a signed trust package, define roles and expiry, and support revocation.

Before work, verify that the device ID, key ID, organization, operator, roles, validity period, and trust-package signature match the deployment record. `UNVERIFIED`, `LOCALLY_TRUSTED`, `ORGANIZATION_VERIFIED`, and `REVOKED` are protocol identifiers; only organization policy may grant authority. Reject an expired, unknown, mismatched, or revoked package.

4. Language selection

Tap the persistent globe/language icon in the top header and choose Deutsch, English, አማርኛ, or ኩኒዲኛ. The selection persists locally. Identifiers, hashes, codes, proper names, and protocol values remain unchanged; explanatory labels are translated.

5. Offline preparation

Before leaving connectivity:

- charge the device and confirm sufficient protected storage;
- open the app, select the correct language, and verify operator/device trust;
- download the assigned work and approved trust package in a production implementation;
- verify date/time, camera, location, and required permissions;
- prepare a safe package-transfer method and a paper fallback;
- never rely on NFC: real chip operations are intentionally blocked.

Offline-ready means local workflows can run. It does not mean identity, trust, backup, or synchronization is complete.

6. Operator authentication

The reference apps do not implement production login. Production must require an organization-issued identity, device unlock, short inactivity timeout, role authorization for every action, and supervisor approval for sensitive corrections or reconciliation. Never share credentials or leave an unlocked device unattended. Stop work if the displayed operator, organization, or role is wrong.

7. Genesis and sack capture

1. Open Workflows/Sacks and choose new batch Genesis.
2. Enter the batch reference and origin exactly as shown on source records.
3. Set the sack count from 1 through 100. Counts outside that range are refused.
4. Create the batch. The app creates deterministic sack references and starts the batch at `UNISSUED`.
5. Compare every displayed sack reference with the physical marking before continuing.

Do not reuse identifiers or silently edit a Genesis event. If source data is wrong, create an authorized correction/void event according to production policy.

8. GPS quality

Capture location outdoors with a clear sky view. Wait for a stable fix and inspect horizontal accuracy, provider, capture time, and mock-location indication where available. The reference acceptance indicator uses 25 metres or better, but deployment policy may be stricter. Retry poor fixes and record why no acceptable fix was possible. GPS is evidence metadata, not proof of origin; never invent, round, or copy coordinates from another site.

9. Lifecycle, controls, damage, opening, and corrections

Normal lifecycle identifiers are `UNISSUED` `ISSUED` `SEALED` `IN_TRANSIT` `OPENED`. `VOID` and `DAMAGED` are terminal exception states. The UI offers only actions allowed from the current state.

- Issue only after checking Genesis and physical sacks.
- Seal only after confirming the approved seal procedure.
- Start transit only when custody and shipment details are correct.
- Open only at the authorized destination and record evidence.
- Mark damaged when a sack or seal is damaged; stop normal processing and follow quarantine policy.
- Mark void when an identifier or record must not be used.
- Never overwrite history. Corrections are new signed events referring to the incorrect record and require the configured authorization.

10. OFFER, ACCEPT, REJECT, and PENDING

An `OFFER` creates a transfer in `PENDING`. Ownership/custody is not complete while pending. The receiver must compare batch/sack references, sender, receiver, condition, and the exact offer hash. `ACCEPT` or `REJECT` must reference that exact hash. A second decision or a mismatched hash must be refused. Keep disputed or concurrent transfers pending/quarantined for supervisor review; do not recreate them to hide the conflict.

11. Photo and document evidence

Photograph labels, seals, condition, and authorized supporting documents clearly. Avoid unrelated people, IDs, screens, or private surroundings. Check focus, completeness, time, batch association, MIME type, size, and SHA-256 reference before continuing. The reference models store metadata and hashes, not a complete secure media vault. Production needs encrypted media storage, upload retry, malware scanning, retention/deletion policy, and access control.

12. Quarantine, conflicts, and review

Quarantine a batch or package when signatures, hashes, sequence, previous hash, identity/trust, lifecycle, evidence, or physical condition cannot be verified. A divergent event branch is not merged automatically. Preserve both branches, record the error code, stop further operational actions, and escalate. “Mark reviewed” only records review in the reference app; production reconciliation requires a signed compensating decision and authorized supervisor.

13. Package export, import, and verification

For export, select create signed export, keep the generated package unchanged, and transfer it through an approved channel. For import, choose import and verify. Verification must finish before any event is accepted: manifest structure/signature, exact file set, safe paths, byte sizes, every file SHA-256, chain root, event chain, and event signatures are checked.

Never partially import, rename/edit package contents, bypass a verification failure, or treat a self-contained signature as organization trust. Invalid or divergent packages stay quarantined. Production also needs size limits, streaming archives, recipient encryption, malware/media scanning, trusted-key lookup, replay controls, and audit-backed synchronization.

14. Error codes and operator response

- SCT-INPUT / GENESIS_SACK_COUNT_INVALID: correct required input or the 1–100 count.
- SCT-STATE / TRANSITION_NOT_ALLOWED: refresh the record and use only an allowed lifecycle action.
- SCT-EVENT: stop and preserve the data; duplicate, sequence, hash, or chain integrity failed.
- SCT-SIGN / KEYSTORE_UNAVAILABLE: stop signing; verify device registration and key protection.
- SCT-PACKAGE / MANIFEST_*: do not import; quarantine the package and report its hash.
- SCT-CONFLICT / IMPORT_QUARANTINED: stop batch processing and request authorized reconciliation.
- SCT-STORAGE: stop work, preserve the device, and contact support; do not clear app data.
- SCT-CHIP / SCT-NFC: do not retry as proof of authenticity; use the approved non-NFC fallback.

Record the exact code, time, device ID, batch/transfer/package identifier, and action. Do not send secrets, complete personal data, APDUs, or chip responses in support messages.

15. Privacy and lost device

Collect only necessary data. Do not place passwords, secret keys, unnecessary personal data, full document images, or precise coordinates in notes, logs, chat, or public systems. Follow consent, access, retention, and deletion policies.

If a device is lost or stolen: report it immediately; revoke the device key and trust package; disable its sessions; identify the last trusted event/package checkpoint; quarantine later unsigned/untrusted activity; remotely lock or erase only under approved policy; and enroll a replacement as a new device. Never restore a signing key by copying an app backup.

16. Troubleshooting

- App will not open: verify supported OS, storage, signed build, and device policy; do not reinstall before support preserves evidence.
- Language is wrong: use the persistent globe icon and restart only if the saved choice is not applied.
- No location: move outdoors, enable permission, wait for accuracy, and record the fallback.
- Import fails: keep the original package unchanged and report its hash plus error code.
- Conflicting history: stop operations and quarantine; never select a branch manually.
- NFC fails: expected with the shipped disabled profile; use the approved fallback.
- Signing/storage fails: stop capture and contact the administrator; do not clear app data or change device time.

17. Mandatory NFC warning

REAL NFC AUTHENTICATION, READ, UPDATE, AND READ-BACK MUST REMAIN BLOCKED until an approved manufacturer-specific proof of concept identifies the chip and memory/application layout; uses authoritative APDUs and status words; defines secure, non-app-embedded key provisioning, diversification, rotation, and revocation; proves iOS/Android compatibility; tests replay, relay, cloning, tearing, interruption, lockout, damage, and read-back; and passes independent security review.

The repository contains no approved production chip profile, APDU set, or provisioning keys. UID is lookup-only and never proof of authenticity.

የSCTracker Native Apps የተጠቃሚ መመሪያ

አማርኛ · የማጣቀሻ ትግበራ

ስሪት 0.1 — መስከረም 2026

አስፈላጊ፡- የiOS እና Android መተግበሪያዎቹ ሊሰሩ የሚችሉ የማጣቀሻ ትግበራዎች ናቸው፤ የምርት ስርዓት፣ የደህንነት ማረጋገጫ፣ የሕግ ምክር ወይም ራስ-ሰር የEUDR ተገዢነት ውሳኔ አይደሉም፡፡

1. ዓላማ እና የደህንነት ገደቦች

SCTracker መሣሪያው ከመስመር ውጭ ሲሆን የቡና ባች፣ ከረጢት፣ ማኅተም፣ ርክክብ፣ አካባቢ፣ የሚዲያ ማጣቀሻ እና የጥቅል ማረጋገጫ ይመዘግባል፡፡ ክስተቶች ሳይሰረዙ ይጨመራሉ እና በሃሽ ይያያዛሉ፡፡

የማጣቀሻው ትግበራ የአካባቢ ፊርማ፣ የሕይወት ዑደት ሕጎች፣ የርክክብ ውሳኔ፣ ለይቶ ማቆያ እና የተረጋገጠ ጥቅል ልውውጥን ያሳያል፡፡ ለምርት የጀርባ ስርዓት ማንነት/ፈቃድ፣ የድርጅት እምነት/ስረዛ፣ የደህንነቱ የተጠበቀ ምዝገባ/መመለስ፣ ምስጢራዊ ክትትል፣ የግላዊነት እና ገለልተኛ የደህንነት ግምገማ ያስፈልጋሉ፡፡ ሃሽ ወይም የመሣሪያ ፊርማ መረጃው እውነት ወይም ተጠቃሚው ፈቃድ እንዳለው አያረጋግጥም፡፡

2. መጫን እና ማዋቀር

Android

1. Android 8.0/API 26 ወይም ከዚያ በላይ እና የማያ መቆለፊያ ይጠቀሙ፤ በድርጅቱ የተፈቀደ የተፈረመ build ብቻ ይጫኑ፡፡
2. ለልማት mobile-native\androidን በAndroid Studio፣ JDK 17 እና Android SDK 35 ይክፈቱ ወይም .\gradlew.bat assembleDebug ያስኪዱ፡፡
3. አስፈላጊውን የካሜራ፣ አካባቢ፣ ፋይል እና NFC ፈቃድ ብቻ ይስጡ፡፡ ይህ ማጣቀሻ የሚዲያ/GPS ሞዴሎችን ይዟል ነገር ግን ሁሉንም የመያዝ ሂደት አላጠናቀቀም፡፡
4. በራስጌው የከመስመር ውጭ ዝግጁነት እና ቋሚ የቋንቋ ምልክት ይፈትሹ፡፡

iOS

1. iOS 17 ወይም ከዚያ በላይ እና የመሣሪያ የይሌፍ ቁጥር ይጠቀሙ፤ የተፈቀደ የተፈረመ build ብቻ ይጫኑ፡፡
2. ለልማት mobile-native\ios\SCTrackerNative.xcodeprojን በXcode 16 ወይም ከዚያ በላይ ክፈቱ እና SCTrackerNative scheme ይምረጡ፡፡
3. በእውነተኛ iPhone NFC ለመጠቀም ትክክለኛ App ID እና NFC Tag Reading entitlement ያስፈልጋል፡፡ Simulator ለልማት ብቻ የሶፍትዌር ቁልፍ ይጠቀማል፡፡
4. የዓለም ኳስ ምልክቱ በሁሉም ገጽ ራስጌ ላይ እንዳለ ያረጋግጡ፡፡

የsimulator መረጃ፣ የልማት ቁልፍ፣ debug build ወይም የምሳሌ ተጠቃሚ ለምርት መዝገብ አይጠቀሙ፡፡

3. የመሣሪያ ምዝገባ እና የእምነት ጥቅል

በመተግበሪያው ያሉ ተጠቃሚዎች፣ ሚናዎች እና የእምነት ሁኔታዎች ቦታ ያዡት ናቸው፡፡ በምርት አስተዳዳሪው መሣሪያውን ይመዘግባል፣ የሕዝብ ቁልፉን ከድርጅትና ከተሰየመ ኦፕሬተር ጋር ያስራል፣ የመሣሪያ/OS/app ስሪትና የቁልፍ ጥበቃን ይመዘግባል፣ ሚናና የማብቂያ ጊዜ ያለው የተፈረመ የእምነት ጥቅል ይሰጣል፣ ስረዛንም ይደግፋል፡፡

ሥራ ከመጀመርዎ በፊት የመሣሪያ ID፣ የቁልፍ ID፣ ድርጅት፣ ኦፕሬተር፣ ሚና፣ ጊዜ እና ፊርማ ይፈትሹ፡፡ UNVERIFIED፣ LOCALLY_TRUSTED፣ ORGANIZATION_VERIFIED እና REVOKED የፕሮቶኮል መለያዎች ናቸው፤ ፈቃድ የሚሰጠው የድርጅት ፖሊሲ ብቻ ነው፡፡ ጊዜው ያለፈ፣ የማይታወቅ፣ የማይዛመድ ወይም የተሻረ ጥቅል ውድቅ ያድርጉ፡፡

4. ቋንቋ መምረጥ

በላይኛው ራስጌ ያለውን ቋሚ የዓለም ኳስ/ቋንቋ ምልክት ይጎብኙ እና Deutsch፣ English፣ አማርኛ ወይም ትግርኛ ይምረጡ። ምርጫው በመሣሪያው ላይ ይቀመጣል። መለያዎች፣ ሃሾች፣ ኮዶች፣ የቦታ/ሰው ስሞች እና የፕሮቶኮል አሴቶች አይተረጎሙም፤ ገላጭ ጽሑፍ ይተረጎማል።

5. ከመስመር ውጭ ዝግጅት

- መሣሪያውን ይጡሉ እና በቂ የተጠበቀ ማከማቻ ያረጋግጡ።
- Appን ይክፈቱ፣ ቋንቋ ይምረጡ፣ የአፕሬተር/መሣሪያ እምነት ይፈትሹ።
- በምርት የተመደበውን ሥራ እና የተፈቀደውን የእምነት ጥቅል ያውርዱ።
- ቀን/ሰዓት፣ ካሜራ፣ አካባቢ እና ፈቃዶችን ይፈትሹ።
- ደህንነቱ የተጠበቀ የጥቅል ማስተላለፊያ እና የወረቀት አማራጭ ያዘጋጁ።
- NFC ላይ አይተማመኑ፤ እውነተኛ የቼፕ ስራዎች ሆነው ታግደዋል።

“ከመስመር ውጭ ዝግጅት” ማለት የአካባቢ ሂደት ይሰራል ማለት ብቻ ነው፤ ማንነት፣ እምነት፣ ምትኬ እና ማመሳሰል ተጠናቋል ማለት አይደለም።

6. የአፕሬተር ማረጋገጫ

የማጣቀሻ app የምርት login የለውም። ምርት በድርጅት የተሰጠ ማንነት፣ የመሣሪያ መክፈቻ፣ አጭር የእንቅስቃሴ ማቆሚያ፣ በሚና የተመሠረተ ፈቃድ እና ለስሱ ማስተካከያ የተቆጣጣሪ ፈቃድ ሊያስገድድ ይገባል። ማስረጃ አያጋሩ፣ የተከፈተ መሣሪያ አይተው። የታየው አፕሬተር፣ ድርጅት ወይም ሚና ስህተት ከሆነ ሥራ ያቁሙ።

7. Genesis እና ከ1 እስከ 100 ከረጢት መያዝ

1. Workflows/Sacks ይክፈቱ እና አዲስ Genesis ይምረጡ።
2. የባች ማጣቀሻና መነሻን ከምንጭ ሰነድ በትክክል ያስገቡ።
3. የከረጢት ብዛት ከ1 እስከ 100 ያድርጉ፤ ሌላ ቁጥር ይከለክላል።
4. ባቹን ይፍጠሩ። Appው ቋሚ የከረጢት መለያዎችን ይፈጥራል እና ባቹ UNISSUED ይሆናል።
5. እያንዳንዱን የማያ መለያ ከእውነተኛ ከረጢቱ ምልክት ጋር ያዛምዱ።

መለያ እንደገና አይጠቀሙ ወይም Genesisን በድብቅ አያርትዱ። ስህተት ካለ በምርት ፖሊሲ መሠረት ፈቃድ ያለው የማስተካከያ/VOID ክስተት ይፍጠሩ።

8. የGPS ጥራት

ከቤት ውጭ ሰማይ በግልጽ ቦታ ይያዙ። የተረጋጋ fix ይጠብቁ እና አግድም ትክክለኛነት፣ provider፣ ጊዜ እና mock-location ምልክትን ይፈትሹ። ማጣቀሻው 25 ሜትር ወይም የተሻለን ይቀበላል፤ የምርት ፖሊሲ የበለጠ ጥብቅ ሊሆን ይችላል። ደካማ መለኪያን ይድገሙ እና ያልተገኘውን ያብራሩ። GPS ማስረጃ መረጃ እንጂ የመነሻ ማረጋገጫ አይደለም፤ ኮርዲኔት አይፍጠሩ፣ አያጠጋጉ ወይም ከሌላ ቦታ አይቅዱ።

9. የሕይወት ዑደት፣ ቁጥጥር፣ ጉዳት፣ መክፈት እና ማስተካከያ

መደበኛ ቅደም ተከተል UNISSUED ISSUED SEALED IN_TRANSIT OPENED ነው። VOID እና DAMAGED የመጨረሻ ልዩ ሁኔታዎች ናቸው።

- Genesisና ከረጢቶችን ከፈተሹ በኋላ Issue ያድርጉ።
- የተፈቀደውን የማሳተም ሂደት ከፈተሹ በኋላ Seal ያድርጉ።
- ርክክብ/ጭነት መረጃ ትክክል ሲሆን ብቻ transit ይጀምሩ።

- በተፈቀደው መድረሻ ብቻ Open ያድርጉ እና ማስረጃ ይመዝግቡ።
- ከረጢት/ማሳተም ከተጎዳ DAMAGED ያድርጉ፣ ስራ ያቁሙ እና ለይቶ ማቆያ ይከተሉ።
- ጥቅም ላይ የማይውል መለያ VOID ያድርጉ።
- ታሪክን አይተኩ፤ ማስተካከያ የተሳሳተውን መዝገብ የሚጠቅስ አዲስ የተፈረመ ክስተት ነው።

10. OFFER፣ ACCEPT፣ REJECT እና PENDING

OFFER ርክክቡን PENDING ያደርጋል፤ በዚህ ጊዜ ጥበቃ/ባለቤትነት አልተላለፈም። ተቀባዩ የባች/ከረጢት መለያ፣ ላኪ፣ ተቀባይ፣ ሁኔታ እና ትክክለኛ offer hash ይፈትሻል። ACCEPT ወይም REJECT ያንኑ ሃሽ መጥቀስ አለበት። ሁለተኛው ሰዓዝ ወይም የማይዛመድ ሃሽ ይከለክላል። ግጭት ወይም ተመሳሳይ ርክክብ ሲኖር ለተቆጣጣሪ ግምገማ በPENDING/ለይቶ ማቆያ ይቆይ።

11. የይቶ እና ሰነድ ማስረጃ

መለያ፣ ማሳተም፣ ሁኔታ እና የተፈቀዱ ሰነዶችን ግልጽና ሙሉ አድርገው ያንሱ። ያልተያያዙ ሰዎች፣ ID፣ ስክሪን ወይም የግል አካባቢ አይካተት። focus፣ ሙሉነት፣ ጊዜ፣ የባች ግንኙነት፣ MIME type፣ መጠን እና SHA-256 ይፈትሹ። ማጣቀሻው ሜታዳታና ሃሽ ይይዛል እንጂ ሙሉ የተጠበቀ ሚዲያ ማከማቻ አይደለም። ምርት ምስጢራ፣ retry፣ malware scan፣ የማቆያ/ስረዛ ፖሊሲ እና መዳረሻ ቁጥጥር ይፈልጋል።

12. ለይቶ ማቆያ፣ ግጭት እና ግምገማ

ፊርማ፣ ሃሽ፣ sequence፣ previous hash፣ ማንነት/እምነት፣ የሕይወት ዑደት፣ ማስረጃ ወይም አካላዊ ሁኔታ ማረጋገጥ ካልተቻለ ባች/ጥቅሉን ለይተው ያቆዩ። የተለያዩ የክስተት ቅርንጫፎች ራስ-ሰር አይቀላቀሉም። ሁለቱንም ያቆዩ፣ የስህተት ኮድ ይመዝግቡ፣ ሥራ ያቁሙ እና ያሳውቁ። “ተገምግሟል” ማለት በማጣቀሻው ግምገማ ብቻ ነው፤ ምርት ፈቃድ ያለው የተቆጣጣሪ የተፈረመ የማስተካከያ ውሳኔ ይፈልጋል።

13. ጥቅል መላክ፣ ማስገባት እና ማረጋገጥ

ለመላክ “የተፈረመ export” ይምረጡ፣ ጥቅሉን አይቀይሩ እና በተፈቀደ መንገድ ያስተላልፉ። ለማስገባት “import and verify” ይምረጡ። ከመቀበል በፊት manifest ቅርጽ/ፊርማ፣ ትክክለኛ ፋይሎች፣ ደህንነቱ የተጠበቀ path፣ byte size፣ ሁሉም SHA-256፣ chain root፣ event chain እና signature ይፈትሻሉ።

ከፊል አያስገቡ፣ ፋይል አይቀይሩ፣ ስህተትን አይዘለሉ፣ የጥቅል ፊርማንም እንደ ድርጅት እምነት አይቁጠሩ። የማይሰራ/የሚጋጭ ጥቅል በለይቶ ማቆያ ይቆይ። ምርት የመጠን ገደብ፣ streaming archive፣ recipient encryption፣ malware/media scan፣ trusted-key lookup፣ replay control እና audit sync ይፈልጋል።

14. የስህተት ኮዶች እና የአፕሬትሪ ምላሽ

- SCT-INPUT / GENESIS_SACK_COUNT_INVALID፡- የጎደለ ግቤት ወይም 1-100 ብዛት ያስተካክላል።
- SCT-STATE / TRANSITION_NOT_ALLOWED፡- መዝገቡን ያድሱ እና የተፈቀደ እርምጃ ብቻ ይውሰዱ።
- SCT-EVENT፡- ያቁሙ እና መረጃውን ያቆዩ፤ duplicate፣ sequence፣ hash ወይም chain ችግኝ አለ።
- SCT-SIGN / KEYSTORE_UNAVAILABLE፡- ፊርማ ያቁሙ፤ ምዝገባና የቁልፍ ጥበቃ ይፈትሹ።
- SCT-PACKAGE / MANIFEST_*፡- አያስገቡ፤ ጥቅሉን ለይተው ያቆዩ እና hash ያሳውቁ።
- SCT-CONFLICT / IMPORT_QUARANTINED፡- የባች ሥራ ያቁሙ እና ፈቃድ ያለው እርቅ ይጠይቁ።
- SCT-STORAGE፡- ሥራ ያቁሙ፤ መሣሪያውን ያቆዩ፤ app data አይሰርዙ።
- SCT-CHIP / SCT-NFC፡- እንደ እውነተኛነት ማስረጃ አይድገው፤ የተፈቀደ non-NFC አማራጭ ይጠቀሙ።

ትክክለኛ ኮድ፣ ጊዜ፣ የመሣሪያ ID፣ የባች/ርክክብ/ጥቅል ID እና እርምጃ ይመዝግቡ። ሚስጥር፣ ሙሉ የግል መረጃ፣ APDU ወይም የቺፕ ምላሽ ለsupport አይላኩ።

15. ግላዊነት እና የጠፋ መሣሪያ

አስፈላጊ መረጃ ብቻ ይሰብስቡ። password፣ secret key፣ የማያስፈልግ የግል መረጃ፣ ሙሉ የሰነድ ምስል ወይም ትክክለኛ coordinate በnote፣ log፣ chat ወይም ህዝባዊ ስርዓት አያስቀምጡ።

መሣሪያው ከጠፋ/ከተሰረቀ ወዲያውኑ ያሳውቁ፤ የመሣሪያ ቁልፍና የእምነት ጥቅል ይሰረዙ፤ session ይዝጉ፤ የመጨረሻ የታመነ checkpoint ይለዩ፤ ቀጣይ ያልታመነ እንቅስቃሴን ለይተው ያቆዩ፤ በፖሊሲ ብቻ remote lock/erase ያድርጉ። አዲስ መሣሪያ እንደ አዲስ ይመዝገብ፤ የፊርማ ቁልፍ ከbackup አይቅዱ።

16. ቶግኝ መፍትሔ

- App ካልተከፈተ፡- OS፣ storage፣ signature እና device policy ይፈትሹ፤ ማስረጃ ከመጠበቅ በፊት reinstall አያድርጉ።
- ቋንቋ ከተሳሳተ፡- ቋሚ globe icon ይጠቀሙ።
- አካባቢ ካልተገኘ፡- ወደ ውጭ ይውጡ፣ permission ይክፈቱ፣ accuracy ይጠብቁ እና fallback ይመዝግቡ።
- Import ካልሰራ፡- ዋናውን ጥቅል ሳይቀይሩ ያቆዩ፣ hash እና code ያሳውቁ።
- History ከተጋጨ፡- ሥራ ያቁሙ እና quarantine ያድርጉ፤ ቅርንጫፍ በእጅ አይምረጡ።
- NFC ካልሰራ፡- በተላከው disabled profile የሚጠበቅ ነው፤ የተፈቀደ fallback ይጠቀሙ።
- Signing/storage ካልሰራ፡- ሥራ ያቁሙ፣ app data አይሰርዙ እና የመሣሪያ ሰዓት አይቀይሩ።

17. አስገዳጅ የNFC ማስጠንቀቂያ

እውነተኛ የNFC ማንነት ማረጋገጥ፣ READ፣ UPDATE እና READ-BACK መታገድ አለበት፤ የተፈቀደ የአምራች-ተኮር PoC ቺፕንና memory/application layoutን እስኪለይ፣ አፊሴላዊ APDU/status word እስኪጠቀም፣ በapp ውስጥ ያልተቀመጠ secure key provisioning/diversification/rotation/revocation እስኪወስን፣ iOS/Android compatibility እስኪያረጋግጥ፣ replay፣ relay፣ cloning፣ tearing፣ interruption፣ lockout፣ damage እና read-back እስኪፈትሽ እና ገለልተኛ የደህንነት ግምገማ እስኪያልፍ ድረስ።

Repositoryው የተፈቀደ የምርት chip profile፣ APDU set ወይም provisioning key የለውም። UID ለፍለጋ ብቻ ነው፤ የእውነተኛነት ማስረጃ አይደለም።

መምርሒ ተጠቃሚ SCTracker Native Apps

ትግርኛ · መወከሲ ትግበራ

ስሪት 0.1 — መስከረም 2026

አገዳሲ፡- ናይ iOSን Androidን መተግበሪታት ዝሰርሑ መወከሲ ትግበራታት እዮም፤ ናይ ምርት ስርዓት፤ ምስክር ውሕስነት፤ ሕጋዊ ምክሪ ወይ ኣውቶማቲክ EUDR ውሳኔ ኣይኮኑን።

1. ዕላማን ደረት ውሕስነትን

SCTracker መሳርሒ ካብ መስመር ወጻኢ ከሎ መረጋገጺ ባች ቡን፤ ከረጺት፤ ማሕተም፤ ርኽክብ፤ ቦታ፤ መወከሲ ሚድያን ምርግጋጽ ጥቕልን ይምዝግብ። ፍጻሜታት ከይተደምሰሱ ይውሰኹን ብሃሽ ይተኣሳስሩን።

እቲ መወከሲ ናይ ከባቢ ፊርማ፤ ሕጊ ዑደት ህይወት፤ ውሳኔ ርኽክብ፤ ፍሉይ ምዕቃብን ዝተረጋገጸ ልውውጥ ጥቕልን የርኢ። ንምርት ድሕረ-ስርዓት መንነት/ፍቓድ፤ ናይ ውድብ እምነት/ስረዛ፤ ውሑስ ምዝገባ/ምምላስ፤ ምስጢር፤ ክትትል፤ ግላውነትን ናጻ ግምገማ ውሕስነትን የድሊ። ሃሽ ወይ ፊርማ መሳርሒ ሓበሬታ ሓቂ ወይ ተጠቃሚ ፍቓድ ከምዘለዎ ኣየረጋግጽን።

2. ምትካልን ምድላውን

Android

1. Android 8.0/API 26 ወይ ልዕሊኡ ምስ screen lock ተጠቐሙ፤ ብውድብ ዝተፈቐደ ዝተፈረመ build ጥራይ ተኸሉ።
2. ንልምዓት `mobile-native\android` ብAndroid Studio፤ JDK 17ን Android SDK 35ን ክፈቱ ወይ `.\gradlew.bat assembleDebug` ኣካይዱ።
3. ዘድሊ ናይ camera፤ location፤ fileን NFCን ፍቓድ ጥራይ ሃቡ። እዚ መወከሲ ሞዴል ሚድያ/GPS ኣለዎ፤ ኩሉ መስርሕ ምቕራጽ ግን ኣይወድኡን።
4. ኣብ header “offline ready”ን ቀዋሚ ምልክት ቋንቋን ከምዘሎ ኣረጋግጹ።

iOS

1. iOS 17 ወይ ልዕሊኡ ምስ passcode ተጠቐሙ፤ ዝተፈቐደ ዝተፈረመ build ጥራይ ተኸሉ።
2. ንልምዓት `mobile-native\ios\SCTrackerNative.xcodeproj` ኣብ Xcode 16 ወይ ልዕሊኡ ክፈቱ፤ SCTrackerNative scheme ምረጹ።
3. NFC ኣብ ናይ ሓቂ iPhone ቅኑስ App IDን NFC Tag Reading entitlementን የድሊ። Simulator ንልምዓት ጥራይ ናይ software መፍትሕ ይጥቀም።
4. ምልክት ዓለም ኣብ ኩሉ header ገጽ ከምዘሎ ኣረጋግጹ።

ሓበሬታ simulator፤ መፍትሕ ልምዓት፤ debug build ወይ ኣብነታዊ ተጠቃሚ ንመዝገብ ምርት ኣይትጠቐሙ።

3. ምዝገባ መሳርሒን ጥቕል እምነትን

ኣብ app ዘለዉ ተጠቀምቲ፤ ተራታትን ኩነታት እምነትን መትሓዚ ቦታ እዮም። ኣብ ምርት ኣመሓዳሪ መሳርሒ ይምዝግብ፤ public key ምስ ውድብን ስም ዘለዎ ኦፕሬተርን የተኣሳስር፤ hardware/OS/appን ደረጃ ሓለዎ መፍትሕን ይምዝግብ፤ ተራን ግዜ ምውዳእን ዘለዎ ዝተፈረመ trust package ይህብ፤ ስረዛ ይድግፍ።

ቅድሚ ስራሕ device ID፤ key ID፤ ውድብ፤ ኦፕሬተር፤ ተራ፤ ግዜን ፊርማን ኣረጋግጹ። UNVERIFIED፤ LOCALLY_TRUSTED፤ ORGANIZATION_VERIFIEDን REVOKEDን መለለዪታት ፕሮቶኮል እዮም፤ ስልጣን ናይ ውድብ ፖሊሲ ጥራይ ይህብ። ግዜኡ ዝሓለፈ፤ ዘይፍለጥ፤ ዘይሰማማዕ ወይ ዝተሰረዘ ጥቕል ንጸጉ።

4. ምርጫ ቋንቋ

አብ ላዕላዊ header ዘሎ ቀዋሚ globe/language icon ጠውቑ፡ Deutsch፣ English፣ አማርኛ ወይ ትግርኛ ምረጹ። ምርጫ አብ መሳርሒ ይዕቀብ። መለለዪታት፣ ሃሽ፣ ኮድ፣ ስማት ሰብ/ቦታን ክብሪ ፕሮቶኮልን ኣይትርጎሙን፤ መግለጺ ጽሑፍ ይትርጎም።

5. ምድላው ካብ መስመር ወጻኢ

- መሳርሒ ምሉእ ቻርጅን እኹል ውሑስ storageን ኣረጋግጹ።
- App ክፈቱ፣ ቋንቋ ምረጹ፣ እምነት ኦፕሬተር/መሳርሒ ኣረጋግጹ።
- ኣብ ምርት ዝተመደበ ስራሕን ዝተፈቓደ trust packageን ኣውርዱ።
- ዕለት/ሰዓት፣ camera፣ locationን permissionsን ኣረጋግጹ።
- ውሑስ መንገዲ ምትሕልላፍ ጥቕልን ናይ ወረቐት fallbackን ኣዳልዉ።
- NFC ኣይትእመኑ፤ ናይ ሓቂ ስርሒት chip ብዕላማ ተዓጽዩ ኣሎ።

“Offline ready” ናይ ከባቢ መስርሕ ይሰርሕ ማለት ጥራይ እዩ፤ መንነት፣ እምነት፣ backupን syncን ወዲኡ ማለት ኣይኮነን።

6. ምርግጋጽ ኦፕሬተር

መወከሲ app ናይ ምርት login የብሉን። ምርት ብውድብ ዝተዋህበ መንነት፣ device unlock፣ ሓጺር inactivity lock፣ role authorization ንኹሉ ተግባርን supervisor approval ንስራሕ correction/reconciliationን ክገድድ ኣለዎ። credentials ኣይትካፈሉ፣ ዝተኸፍተ መሳርሒ ኣይትግደፉ። ዝርእ ኦፕሬተር፣ ውድብ ወይ ተራ ጌጋ እንተኾይኑ ስራሕ ኣቋርጹ።

7. Genesisን 1 ክሳብ 100 ከረጹት ምቕራጽን

1. Workflows/Sacks ክፈቱ፣ ሓድሽ Genesis ምረጹ።
2. መወከሲ ባችን መበገሲን ካብ ምንጪ ሰነድ ብልክዕ ኣእትዉ።
3. ብዝሒ ከረጹት ካብ 1 ክሳብ 100 ግበሩ፤ ካልእ ቁጽሪ ይንጸግ።
4. ባች ፍጠሩ። App ቀዋሚ መለለዪ ከረጹት ይፈጥር፣ ባች UNISSUED ይኸውን።
5. ነፍሲ ወከፍ ኣብ screen ዘሎ መለለዪ ምስ ኣብ ከረጹት ዘሎ ምልክት ኣዛምዱ።

መለለዪ ዳግማይ ኣይትጠቐሙ፣ Genesis ብሕቡእ ኣይትቐይሩ። ጌጋ እንተሎ ብፖሊሲ ምርት ዝተፈቓደ correction/VOID event ፍጠሩ።

8. ጽሬት GPS

ኣብ ደገ ንሰማይ ግሉጽ ርእይቶ ዘለዎ ቦታ ቅረጹ። ርጉእ fix ተጸብዩ፣ horizontal accuracy፣ provider፣ capture timeን mock-location signalን መርምሩ። መወከሲ 25 ሜትር ወይ ዝሓሸ ይቕበል፤ ፖሊሲ ምርት ዝጸንዐ ክኸውን ይኽእል። ድኹም fix ድገሙ፣ ቅቡል fix ዘይተረኽበሉ ምኽንያት መዝግቡ። GPS መረጋገዲ ሓበሬታ እዩ እምበር መርትዖ መበገሲ ኣይኮነን፤ coordinate ኣይትፍጠሩ፣ ኣይትጠጋግዑ ወይ ካብ ካልእ ቦታ ኣይትቐድሑ።

9. ዑደት ህይወት፣ ቁጽጽር፣ ጉድኣት፣ ምኽፋትን ምእራምን

ንቡር ቅደም UNISSUED ISSUED SEALED IN_TRANSIT OPENED እዩ። VOIDን DAMAGEDን ናይ መወዳእታ ፍሉይ ኩነታት እዮም።

- Genesisን ከረጹታትን ድሕሪ ምርግጋጽ Issue ግበሩ።
- ዝተፈቓደ መስርሕ ማሕተም ድሕሪ ምርግጋጽ Seal ግበሩ።
- ርኽክብ/ምልኣኽ ሓበሬታ ቅኑዕ ክኸውን ከሎ transit ጀምሩ።

- ኣብ ዝተፈቐደ መዕረፊ ጥራይ Open ግበሩ፣ መረጋገጺ መዝግቡ።
- ከረጲት/ማሕተም እንተተገዲኡ DAMAGED ግበሩ፣ ስራሕ ኣቋርጹ፣ quarantine ተኸተሉ።
- ዘይጠቅም መለለዪ VOID ግበሩ።
- ታሪኽ ኣይትደምስሱ፣ correction ነቲ ጌጋ ዝጠቅስ ሓደሽ ዝተፈረመ event እዩ።

10. OFFER፣ ACCEPT፣ REJECTን PENDINGን

OFFER ርኽክብ PENDING ይገብር፣ ኣብዚ ግዜ custody/ownership ኣይተሰጋገረን። ተቐባሊ መለለዪ ባች/ከረጲት፣ ላኡኪ፣ ተቐባሊ፣ ኩነታትን exact offer hashን ይምርምር። ACCEPT ወይ REJECT ነቲ ልክዕ ሃሽ ክጠቅስ ኣለዎ። ካልኣይ ውሳኔ ወይ ዘይሰማማዕ ሃሽ ይንጸግ። ዝተኸራኸረ ወይ concurrent transfer ንግምገማ supervisor ኣብ PENDING/quarantine ይጽናሕ።

11. መረጋገጺ ስእልን ሰነድን

label፣ seal፣ ኩነታትን ዝተፈቐደ ሰነድን ንጹርን ምሉእን ጌርኩም ሰኣሉ። ዘይምልከቶም ሰባት፣ ID፣ screen ወይ ውልቃዊ ከባቢ ኣይተእትዉ። focus፣ completeness፣ time፣ batch association፣ MIME type፣ sizeን SHA-256ን ኣረጋግጹ። መወከሲ metadataን hashን ይሕዝ፣ ምሉእ ውሑስ media vault ኣይኮነን። ምርት encryption፣ upload retry፣ malware scan፣ retention/deletion policyን access controlን የድሊ።

12. ፍሉይ ምዕቃብ፣ ግጭትን ግምገማን

ፊርማ፣ hash፣ sequence፣ previous hash፣ identity/trust፣ lifecycle፣ evidence ወይ physical condition ክረጋገጽ እንተዘይክኣሉ ባች/ጥቕል quarantine ግበሩ። ዝተፈለለ event branch ብኣውቶማቲክ ኣይወሃሃድን። ክልቲኡ ዓቕቡ፣ error code መዝግቡ፣ ስራሕ ኣቋርጹን ኣሕልፉን። “Mark reviewed” ኣብ መወከሲ ግምገማ ጥራይ ይምዝግብ፣ ምርት ብዝተፈቐደ supervisor ዝተፈረመ compensating decision የድሊ።

13. Export፣ importን verification ጥቕልን

ንexport “create signed export” ምረጹ፣ ጥቕል ከይቀየርኩም ብዝተፈቐደ መንገዲ ኣሕልፉ። ንimport “import and verify” ምረጹ። ቅድሚ ምቕባል manifest structure/signature፣ exact file set፣ safe path፣ byte size፣ ኩሉ SHA-256፣ chain root፣ event chainን signatureን ይፍተሽ።

ብኸፊል ኣይተእትዉ፣ file ኣይትቐይሩ፣ verification failure ኣይትሕለፉ፣ package signature ከም organization trust ኣይትቐጽሩ። Invalid/divergent package quarantine ይጽናሕ። ምርት size limit፣ streaming archive፣ recipient encryption፣ malware/media scan፣ trusted-key lookup፣ replay controlን audited syncን የድሊ።

14. Error codeን ምላሽ ኦፕሬተርን

- SCT-INPUT / GENESIS_SACK_COUNT_INVALID፡- ዘድሊ input ወይ ብዝሒ 1–100 ኣስተኻኽሉ።
- SCT-STATE / TRANSITION_NOT_ALLOWED፡- record refresh ግበሩ፣ ዝተፈቐደ action ጥራይ ውሰዱ።
- SCT-EVENT፡- ኣቋርጹን data ዓቕቡን፣ duplicate፣ sequence፣ hash ወይ chain ጌጋ ኣሎ።
- SCT-SIGN / KEYSTORE_UNAVAILABLE፡- signing ኣቋርጹ፣ registrationን key protectionን መርምሩ።
- SCT-PACKAGE / MANIFEST_*፡- import ኣይትግበሩ፣ package quarantine ግበሩ፣ hash ሓብሩ።
- SCT-CONFLICT / IMPORT_QUARANTINED፡- batch work ኣቋርጹ፣ authorized reconciliation ሕተቱ።
- SCT-STORAGE፡- ስራሕ ኣቋርጹ፣ መሳርሒ ዓቕቡ፣ app data ኣይትደምስሱ።
- SCT-CHIP / SCT-NFC፡- ከም መረጋገጺ authenticity ኣይትድገሙ፣ approved non-NFC fallback ተጠቐሙ።

Exact code፣ time፣ device ID፣ batch/transfer/package IDን actionን መዝግቡ። secret፣ full personal data፣ APDU ወይ chip response ንsupport ኣይትልኡኹ።

15. ግላውነትን ዝጠፍኦ መሳርሒን

ዘድሊ data ጥራይ ኣክቡ፡ password፣ secret key፣ ዘየድሊ personal data፣ full document image ወይ exact coordinate ኣብ note፣ log፣ chat ወይ public system ኣይተቐምጡ፡፡

መሳርሒ ኣንተጠፊኡ/ተሰሪቐ ብኡነብኡ ሓብሩ፤ device keyን trust packageን revoke ግበሩ፤ sessions disable ግበሩ፤ last trusted event/package checkpoint ፍለጡ፤ ድሕሪኡ ዘሎ untrusted activity quarantine ግበሩ፤ remote lock/erase ብፖሊሲ ጥራይ ግበሩ፡፡ Replacement ከም ሓድሽ device ይመዝገብ፤ signing key ካብ app backup ኣይትቐድሑ፡፡

16. ፍታሕ ጸገም

- App ኣንተዘይተኸፈቱ፡ OS፣ storage፣ signed buildን device policyን መርምሩ፤ ቅድሚኡ evidence preservation reinstall ኣይትግበሩ፡፡
- ቋንቋ ጌጋ ኣንተኸይኑ፡ persistent globe icon ተጠቐሙ፡፡
- Location ኣንተዘይሃልዩ፡ ናብ ደገ ውጹ፣ permission ክፈቱ፣ accuracy ተጸበዩ፣ fallback መዝግቡ፡፡
- Import ኣንተፈሺሉ፡ original package ከይቀየረ ዓቕቡ፣ hashን codeን ሓብሩ፡፡
- History conflict፡ ስራሕ ኣቋርጹ፣ quarantine ግበሩ፣ branch ብኢድ ኣይትምረጹ፡፡
- NFC ኣንተፈሺሉ፡ ምስ shipped disabled profile ዝጽበ እዩ፣ approved fallback ተጠቐሙ፡፡
- Signing/storage failure፡ capture ኣቋርጹ፣ app data ኣይትደምስሱ፣ device time ኣይትቐይሩ፡፡

17. ግዴታዊ መጠንቀቓታ NFC

ናይ ሓቂ NFC AUTHENTICATION፣ READ፣ UPDATEን READ-BACKን ተዓጽዩ ክጸንሕ ኣለዎ፤ ዝተፈቐደ manufacturer-specific PoC ነቲ chipን memory/application layoutን ክሳብ ዝፈሊ፣ authoritative APDU/status word ክሳብ ዝጥቀም፣ ኣብ app ዘይተቐመጠ secure key provisioning/diversification/rotation/revocation ክሳብ ዝውስን፣ iOS/Android compatibility ክሳብ ዘረጋግጽ፣ replay፣ relay፣ cloning፣ tearing፣ interruption፣ lockout፣ damageን read-backን ክሳብ ዝፍትንን ናጻ security review ክሳብ ዝሓልፍን፡፡

Repository ዝተፈቐደ production chip profile፣ APDU set ወይ provisioning key የብሉን፡፡ UID ንlookup ጥራይ እዩ፤ መረጋገጺ authenticity ኣይኮነን፡፡